CARTER-BEAU BENSON, PhD

Project Manager & Ontologist | Applied Ontology | Knowledge Representation | Semantic Interoperability
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PROFILE

Project Manager and Ontologist at CUBRC with a PhD in Philosophy from the University at Buffalo. Specializes in applied ontology, BFO/CCO-based modeling, knowledge representation, semantic interoperability, knowledge graphs, ontology-supported AI, and the management of ontology-driven technical projects. Published research spans biomedical ontology, realizable entities, mid-level ontology architecture, large language models, and sports ontology.

TECHNICAL EXPERTISE

Semantic technologies	OWL 2, RDF, RDFS, SPARQL, SHACL, RML
Ontology frameworks	Basic Formal Ontology (BFO), Common Core Ontologies (CCO), OBO Foundry
Logic	First-order logic, description logic, modal logic
Tools & platforms	Protégé, TopBraid Composer, Apache Jena Fuseki/TDB2, Apache NiFi, Git/GitHub, Jira, Visual Studio Code

PROFESSIONAL EXPERIENCE

Project Manager and Ontologist — *CUBRC, Inc.* **2024–Present**

Project management, applied ontology, and semantic modeling in support of interoperable, machine-readable data systems and knowledge representation.

Data Science Intern — *Buffalo Manufacturing Works* **2024**
Ontology Project Manager — *ScienceCast* **2023–2024**
Research Assistant — *University at Buffalo — John Beverley* **2023–2024**
Research Assistant — *University at Buffalo — Barry Smith* **2023–2024**

EDUCATION

PhD, Philosophy — *University at Buffalo* [link] **2026**

Dissertation: "Netting Expectations: The Football Ontology." Formal ontology of association football grounded in BFO and CCO, with emphasis on soccer processes, information entities, expected goals (xG), and semantic interoperability.

MA, Philosophy — *University at Buffalo* **2020**
BS — *University at Buffalo* **2017**
AA, Liberal Arts — *SUNY Schenectady* **2012**

PEER-REVIEWED PUBLICATIONS

- Beverley, J., Babcock, S., Benson, C., De Colle, G., Cohen, S., Diehl, A. D., Challa, R. A. N. R., Mavrovich, R. A., Billig, J., Huffman, A., & He, Y. (2025). "A fourfold pathogen reference ontology suite." Journal of Biomedical Semantics, 16, 12. https://doi.org/10.1186/s13326-025-00333-6 Link
- Rabenberg, M., Benson, C., Donato, F., He, Y., Huffman, A., Babcock, S., & Beverley, J. (2024). "Grounding Realizable Entities." CEUR Workshop Proceedings, Vol. 3939. 15th International Conference on Biological and Biomedical Ontology (ICBO 2024). Link
- Beverley, J., De Colle, G., Jensen, M., Benson, C.-B., & Smith, B. (2024). "Middle Architecture Criteria." CEUR Workshop Proceedings, Vol. 3882, pp. 1–12. Joint Ontology Workshops (JOWO 2024). Link
- Beverley, J., Franda, F., Karray, H., Maxwell, D., Benson, C.-B., & Smith, B. (2024). "Ontologies, Arguments, and Large-Language Models." CEUR Workshop Proceedings, Vol. 3882, pp. 1–9. Joint Ontology Workshops (JOWO 2024). Link

Forthcoming

- Benson, C.-B., Sculley, A., Liebers, A., & Beverley, J. "My Ontologist: Evaluating BFO-Based AI for Definition Support." Forthcoming in the Proceedings of Semantic Technologies for Intelligence, Defense, and Security (STIDS 2026). Preprint: arXiv:2407.17657. Link

ONTOLOGY & KNOWLEDGE-GRAPH PROJECTS

BaseballO — *Open-source research system* [link] **2026–Present**

Ontology and knowledge-graph system for querying baseball as connected game events rather than isolated statistics. Uses RML mapping, SHACL validation, SPARQL, Apache Jena Fuseki/TDB2, and Apache NiFi to build and query checked game representations.

Football Ontology (FutOn) — *Association football / soccer* [link] **2023–Present**

BFO- and CCO-grounded ontology developed with the dissertation. Models soccer players, roles, positions, processes, information entities, and expected goals (xG) to support interoperable representation and analysis of soccer data..

SELECTED PRESENTATIONS

My Ontologist: Evaluating BFO-Based AI for Definition Support — *Semantic Technologies for Intelligence, Defense, and Security (STIDS)* **2026**

Alignment of AM-CDM to BFO — *Additive Manufacturing Common Data Model Working Group* **2024**

EOBP Conference Summary — *Department of Defense and Intelligence Community Foundry* **2024**

Modeling Expected Goals — *NCOR Common Core Ontologies Working Group* **2024**

The Importance of Interoperability: A Case Study in DBpedia — *Graduate Philosophy Association Lunchtime Talk* **2023**

Incorporating the OBO Foundry and CCO into FutOn — *University at Buffalo Ontology Working Group — Invited* **2023**

Ontology Workflow at UB — *Institute for Artificial Intelligence and Data Science Poster Presentation — Invited* **2023**

PROFESSIONAL SERVICE & LEADERSHIP

Chair, Common Core Ontologies Working Group — *National Center for Ontological Research* **2023–2024**

Member, Common Core Ontologies Working Group — *National Center for Ontological Research* **2023–Present**

Contributing Member — *IEEE P3195 Common Core Ontologies Working Group* **2023–2025**

Member — *Additive Manufacturing Common Data Model Working Group* **2024–Present**

Member — *Department of Defense and Intelligence Community Foundry* **2023–Present**

Primary Organizer — *Enhanced Object-Based Production Conference, Part 2* **2024**

Assistant Organizer — *BFO Summit Meeting* **2023**

Primary Organizer — *Enhanced Object-Based Production Conference* **2023**

AWARDS

Outstanding Teaching Assistant / Research Assistant Award — *University at Buffalo Department of Philosophy* **2022–2023**

Peter H. Hare Department Citizenship Award — *University at Buffalo Department of Philosophy* **2021–2022**

Peter H. Hare Department Citizenship Award — *University at Buffalo Department of Philosophy* **2020–2021**

Outstanding Teaching Assistant / Research Assistant Award — *University at Buffalo Department of Philosophy* **2019–2020**

SUNY TA/RA Scholarship — *University at Buffalo* **2018–2024**